From Physical to Human Capital Accumulation: Inequality in the Process of Development

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Review of Economic Studies, 2004

Inequality and Growth: A Unified Perspective

- A unified theory of inequality and growth:
  - Captures the changing effect of inequality on growth
  - Reconciles the Traditional and Modern Perspectives
    - Inequality fosters growth driven by physical capital accumulation
    - Inequality retards growth driven by human capital accumulation

The Traditional Perspective

- Inequality promotes economic growth

  Keynes (1919), Kaldor (1957)

  - The marginal propensity to save increases with income
  - Inequality channels resources towards individuals whose marginal propensity to save is higher
    - Inequality $\Rightarrow$ Capital accumulation $\Rightarrow$ Economic growth

The Modern Perspective

- Inequality can hamper economic growth

  Galor and Zeira (1988, 1993)

  - Credit market imperfections
    - Limit human capital investment among individuals with limited resources
  - Inequality generates suboptimal investment in human capital
    - Lower human capital accumulation $\Rightarrow$ Slower economic growth

Reconciliation: Asymmetric Human and Physical Capital Accumulation

- Human capital accumulation is embodied in individuals (physiological constraints)
  - Individual investment in human capital faces diminishing returns
    - Broad-based participation fosters aggregate human capital accumulation
- Physical capital accumulation is disembodied (unconstrained by individual capacities)
  - Returns do not diminish with an individual's holdings of physical capital
    - Wealth concentration fosters capital accumulation (MPS rises with income)

Main Hypothesis

- The transition from physical to human capital accumulation as the prime engine of growth changes the effect of inequality on growth
- Early stages of development:
  - Physical capital accumulation is the prime engine of growth
    - Inequality fosters growth (channeling resources toward individuals with a higher MPS)
- Later stages of development:
  - Capital-skill complementarity raises the return to human capital
  - Human capital accumulation becomes the prime engine of growth
    - Inequality retards growth (limiting human capital investment under credit constraints)

Mechanism: Endogenous Reversal in the Relative Return to Human Capital

- Early stages of growth:
  - Physical capital is scarce
  - Return to PC $>$ Return to HC
    - Inequality fosters economic growth
- Later stages of growth (as physical capital accumulates)
  - Capital-skill complementarity raises the return to human capital
  - Return to HC $>$ Return to PC (for credit-constrained individuals)
    - Inequality hinders economic growth

The Basic Structure of the Model

- An overlapping-generations economy
- Discrete time: $t = 0, 1, 2, 3, \ldots$.
- One good
- Two factors:
  - Physical capital (PC)
  - Human capital (HC)

Production of Final Output

- Output produced at time t:

$$Y_t = F(K_t, H_t) \equiv H_t f(k_t)$$

  - K_t: Physical capital (PC)
  - H_t: Human capital (HC) (measured in efficiency units of labor)
  - $k_t \equiv K_t / H_t$

Factor Prices

- Inverse demand for factors of production at time t

$$\begin{aligned} r_t &= f'(k_t) - \delta \equiv r(k_t) \\ w_t &= f(k_t) - f'(k_t)k_t \equiv w(k_t) \end{aligned}$$

  - r_t: net return to capital
  - w_t: return to an efficiency unit of labor
  - $\delta \in [0, 1]$: depreciation rate

Individuals

- A continuum of measure one in each generation
- Each individual has one parent and one child
- Identical in:
  - Preferences
  - Innate abilities
- Differ in:
  - Parental income $\Rightarrow$ Investment in HC

Individuals of Generation t

- First period of life (period t):
  - Human capital formation (Financed by the parent)
- Second period of life (period $t+1$):
  - Supply their efficiency units of labor and generate wage income
  - Allocate wealth between consumption and intergenerational transfers

Individual i of Generation t: Preferences and Wealth

- Preferences: (non-homothetic; average propensity to bequeath rises with wealth)

$$u_t^i = (1-\beta)\log c_{t+1}^i + \beta \log(\overline{\theta} + b_{t+1}^i)$$

  - c_{t+1}^i – consumption (including child's consumption)
  - b_{t+1}^i – transfers to offspring

- Second-period wealth:

$$\omega_{t+1}^i = w_{t+1} h_{t+1}^i + x_{t+1}^i$$

  - w_{t+1} – wage per efficiency unit of labor
  - h_{t+1}^i – efficiency units of labor
  - x_{t+1}^i – inheritance

Individual i of Generation t: Budget Constraint

- Second-period budget constraint:

$$c^i_{t+1} + b^i_{t+1} \leq \omega^i_{t+1}$$

  - c^i_{t+1} – consumption
  - b^i_{t+1} – transfers to offspring

Individual i of Generation t: Intergenerational Transfers

- Transfer to offspring, b^i_{t+1}, is allocated between:
  - Financing offspring's education – e^i_{t+1}
  - Saving for offspring's future wealth

$$s^i_{t+1} = b^i_{t+1} - e^i_{t+1}$$

- Offspring's inheritance in period $t+2$:

$$x^i_{t+2} = s^i_{t+1} R_{t+2} = (b^i_{t+1} - e^i_{t+1}) R_{t+2}$$

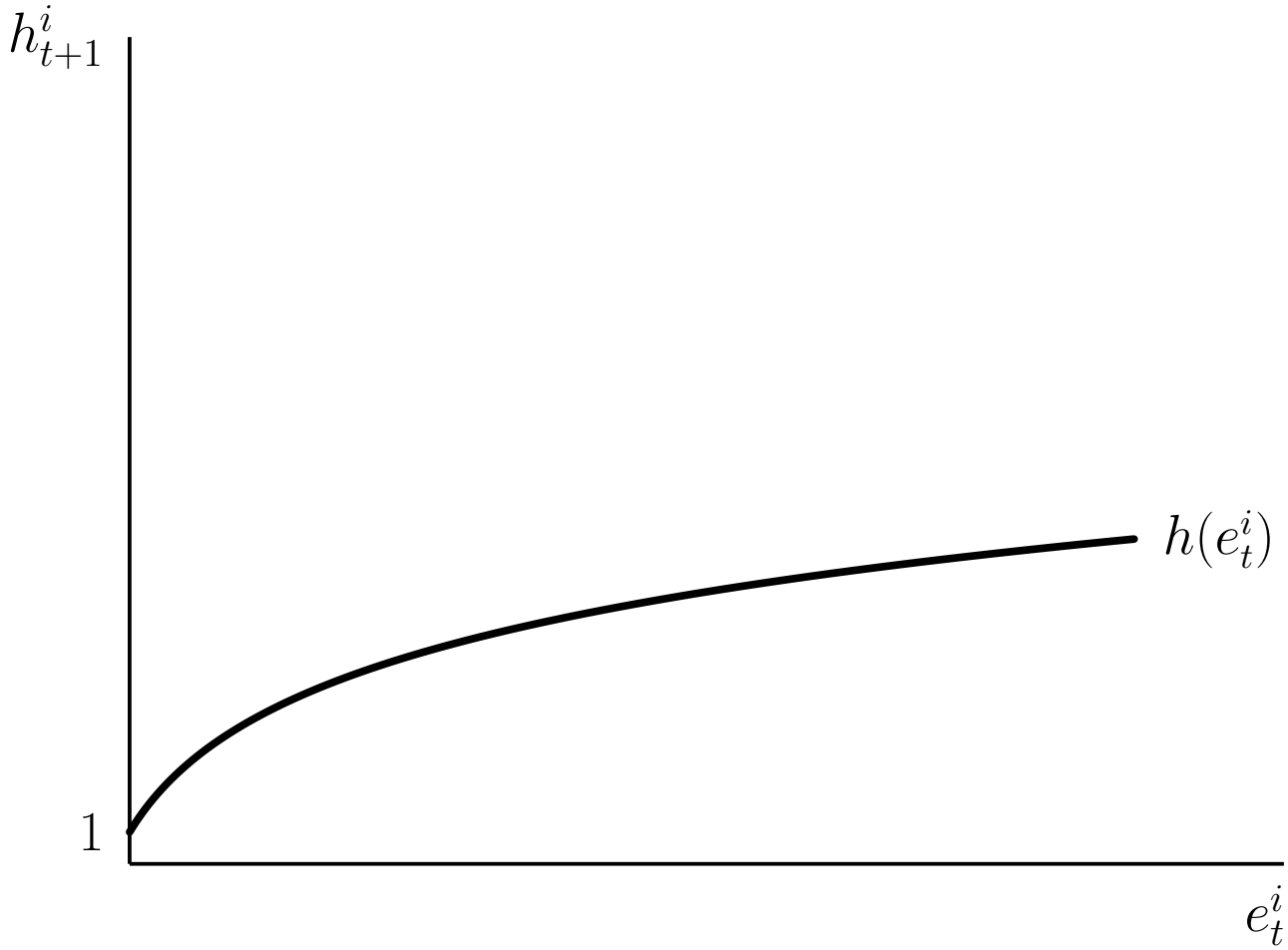
Individual i of Generation t: Human Capital Formation

- Investment in education in period t: e_t^i
- Efficiency units of labor in period $t+1$:

$$h_{t+1}^i = h(e_t^i)$$

  - $h'(e_t^i) > 0, \quad h''(e_t^i) < 0, \quad h(0) > 0, \quad h'(0) < \infty$

Human Capital Formation: Diminishing Returns to Education



Investment in Education: Individual i of Generation t

- Education is chosen to maximize second-period wealth
- In the absence of borrowing constraints:

$$e_t^i = \arg\max \omega_{t+1}^i = \arg\max \left[w_{t+1} h_{t+1}^i + x_{t+1}^i \right]$$

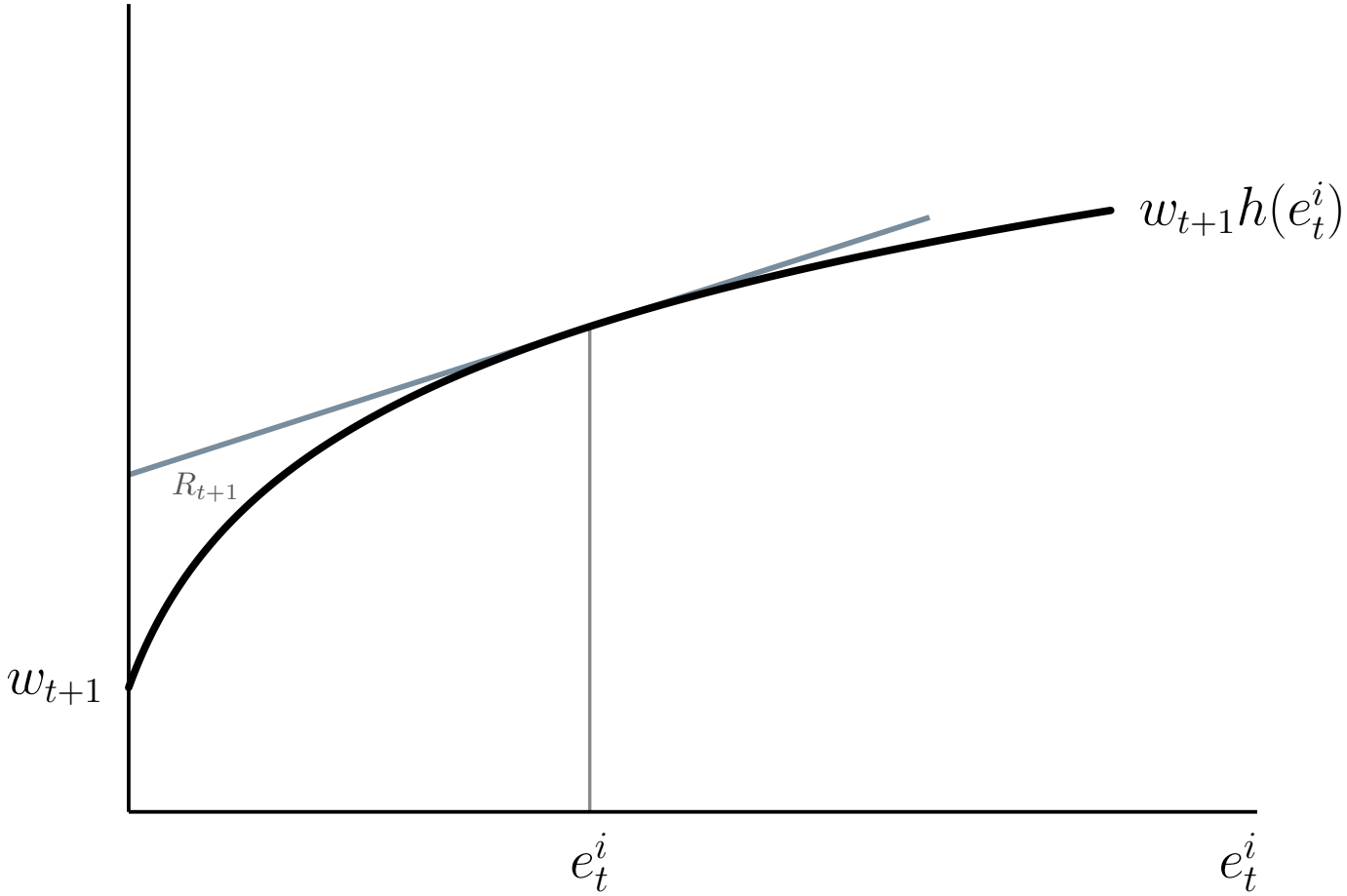
$$= \arg\max \left[w_{t+1} h(e_t^i) + (b_t^i - e_t^i) R_{t+1} \right]$$

  - If $w_{t+1} h'(0) > R_{t+1}$, an interior solution exists

$$w_{t+1} h'(e_t^i) = R_{t+1}$$

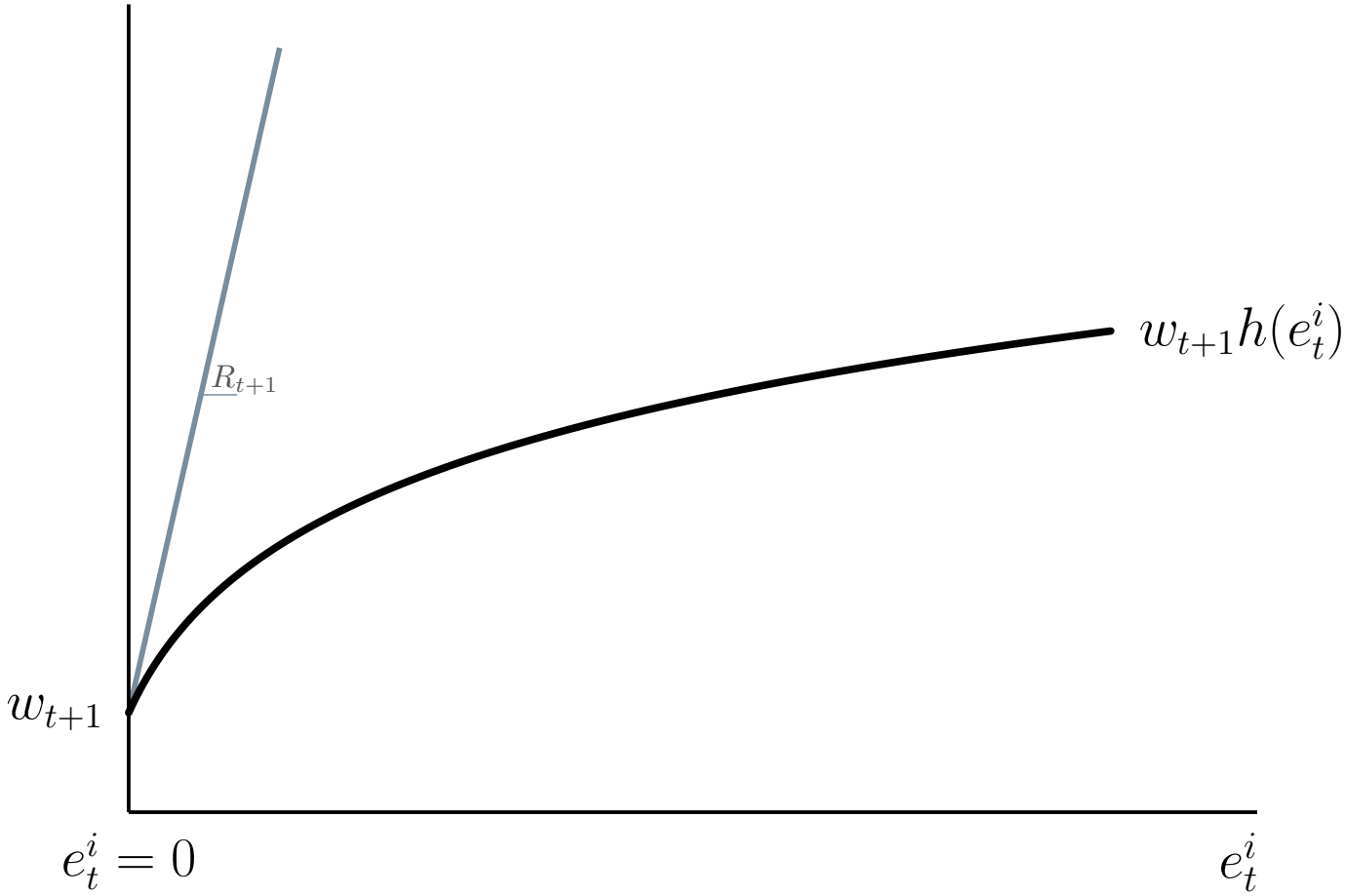
    - Return to saving (R_{t+1}) = return to human capital $(w_{t+1} h'(e_t^i))$
    - $e_t^i = e_t$ for all i

Investment in Education: The Interior Solution

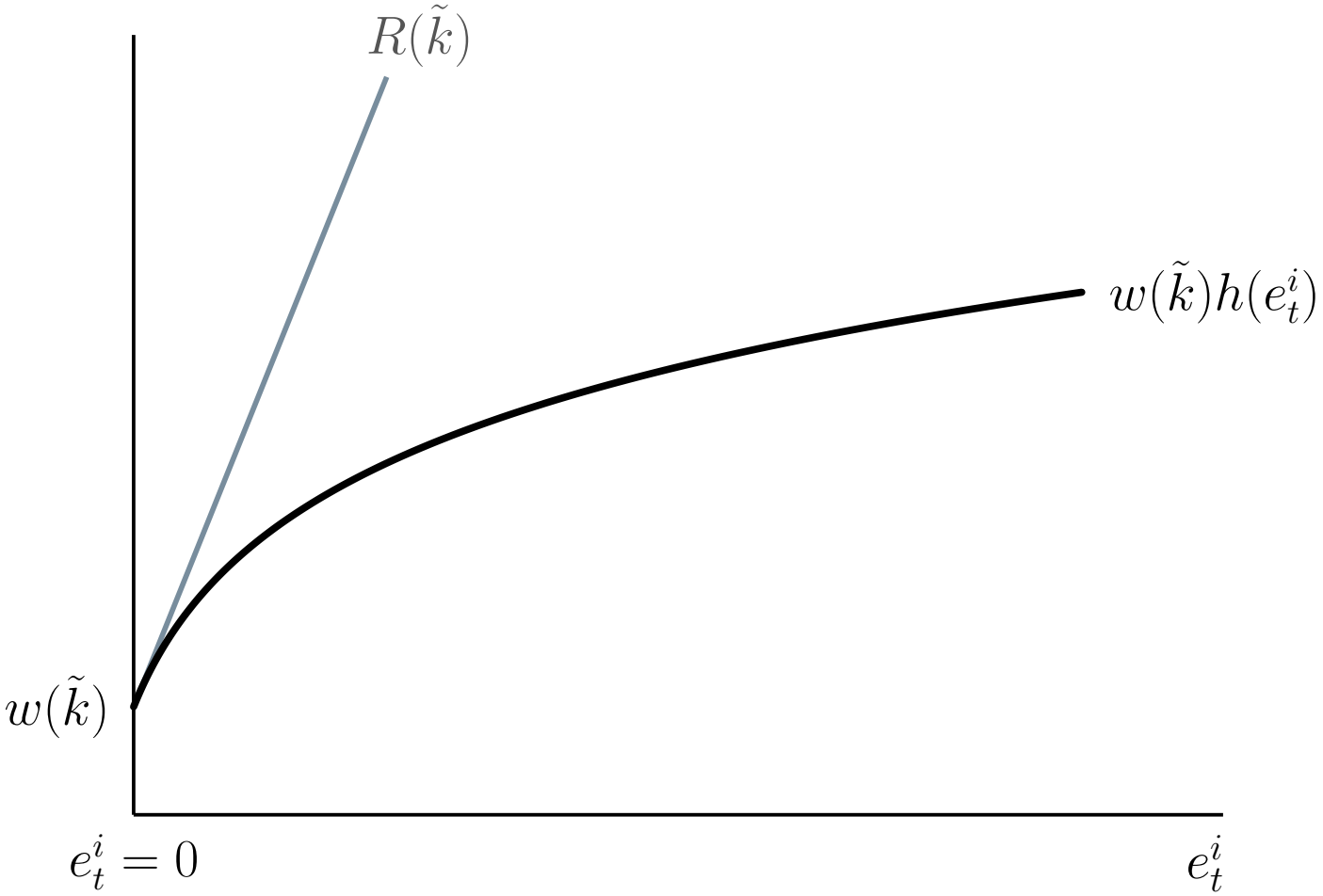


Investment in Education: The Corner Solution

$$R_{t+1} > w_{t+1}h'(0)$$



Investment in Education: The Threshold



Optimal Investment in Education

$$e_t^i = e(k_{t+1}) \begin{cases} = 0 & \text{if } k_{t+1} \leq \widetilde{k}, \\ > 0 & \text{if } k_{t+1} > \widetilde{k}. \end{cases}$$

- Above $\widetilde{k}$ investment in education rises with the capital–labor ratio:

$$e'(k_{t+1}) > 0 \qquad \text{if } k_{t+1} > \widetilde{k}$$

Borrowing Constraint on Investment in Human Capital

- Individuals cannot borrow to finance their offspring's education

$$e_t^i = \min\left[e(k_{t+1}), b_t^i\right]$$

Intergenerational Transfers

- Preferences:

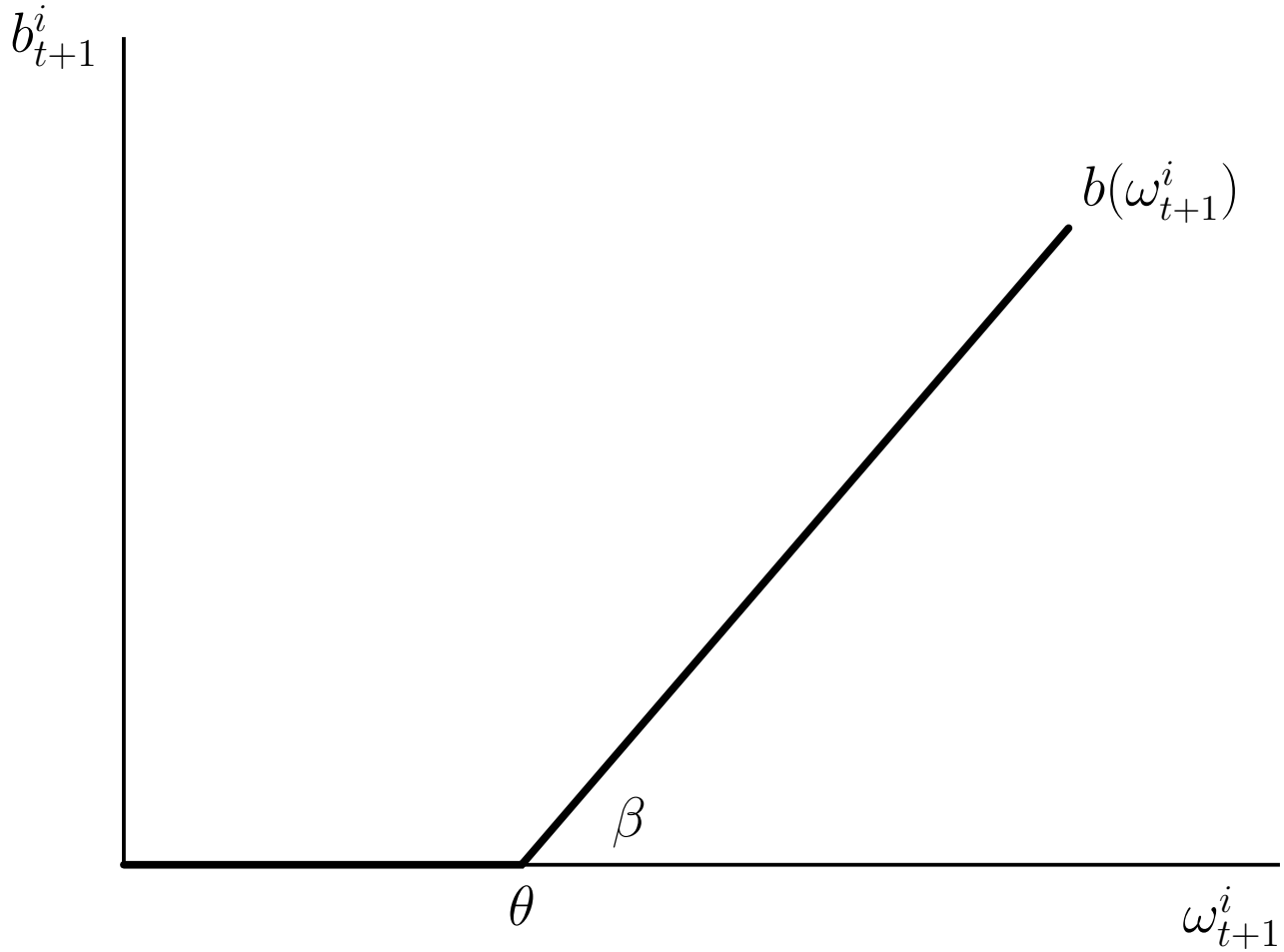
$$u_t^i = (1-\beta)\log c_{t+1}^i + \beta \log(\overline{\theta} + b_{t+1}^i)$$

- Optimal transfer to offspring:

$$b_{t+1}^i = b(\omega_{t+1}^i) \equiv \begin{cases} \beta(\omega_{t+1}^i - \theta) & \text{if } \omega_{t+1}^i > \theta, \\ 0 & \text{if } \omega_{t+1}^i \leq \theta. \end{cases}$$

where $\theta \equiv \overline{\theta}(1-\beta)/\beta$.

Optimal transfer of a member i of generation t



Saving and Wealth

$$s_t^i = \begin{cases} b_t^i & \text{if } k_{t+1} \leq \widetilde{k}, \\ b_t^i - e_t^i & \text{if } k_{t+1} > \widetilde{k}. \end{cases}$$

- The saving rate $s_{t+1}^i / \omega_{t+1}^i$ is nondecreasing in wealth ω_{t+1}^i

Initial Wealth Distribution

The economy consists of two groups in period 0:

- Rich (R)
  - Fraction λ of all adult individuals
  - Own equal shares of the *initial* capital stock
- Poor (P)
  - Fraction $1-\lambda$ of all adult individuals
  - Own no *initial* capital

Factor Accumulation

$$
\begin{aligned}
K_{t+1} &= \int_0^1 s_t^i di = \lambda(b_t^R - e_t^R) + (1-\lambda)(b_t^P - e_t^P) \\
&= K(b_t^R, b_t^p, k_{t+1}) \\
H_{t+1} &= \int_0^1 h_{t+1}^i di = \lambda h(e_t^R) + (1-\lambda) h(e_t^P) \\
&= H(b_t^R, b_t^P, k_{t+1})
\end{aligned}
$$

The Evolution of the Capital-Labor Ratio

$$k_{t+1} = \frac{K_{t+1}}{H_{t+1}} = \frac{K(b_t^R, b_t^P, k_{t+1})}{H(b_t^R, b_t^P, k_{t+1})}$$

$$\Longrightarrow$$

$$k_{t+1} = \kappa(b_t^R, b_t^P)$$

The Evolution of Transfers within group i = R,P

$$b^i_{t+1} = \max\{\beta[w_{t+1}h(e^i_t) + (b^i_t - e^i_t)R_{t+1} - \theta], 0\}$$

$$\Longrightarrow$$

$$b^i_{t+1} = \phi(b^i_t, k_{t+1}) = \phi(b^i_t, \kappa(b^R_t, b^P_t)) \equiv \psi^i(b^R_t, b^P_t)$$

There exists $\widehat{k}$, a critical level of k below which individuals who do not receive parental transfers (i.e., $b^i_t = e^i_t = 0$) do not transfer income to their offspring: $w(\widehat{k}) = \theta$

$$b^i_{t+1} = \phi(0, k_{t+1}) \begin{cases} = 0 & if \quad k_{t+1} \leq \widehat{k} \\ > 0 & if \quad k_{t+1} > \widehat{k} \end{cases}$$

The Dynamical System

- The evolution of transfers in the two groups is governed by:

$$b^P_{t+1} = \psi^P(b^R_t, b^P_t),$$

$$b^R_{t+1} = \psi^R(b^R_t, b^P_t).$$

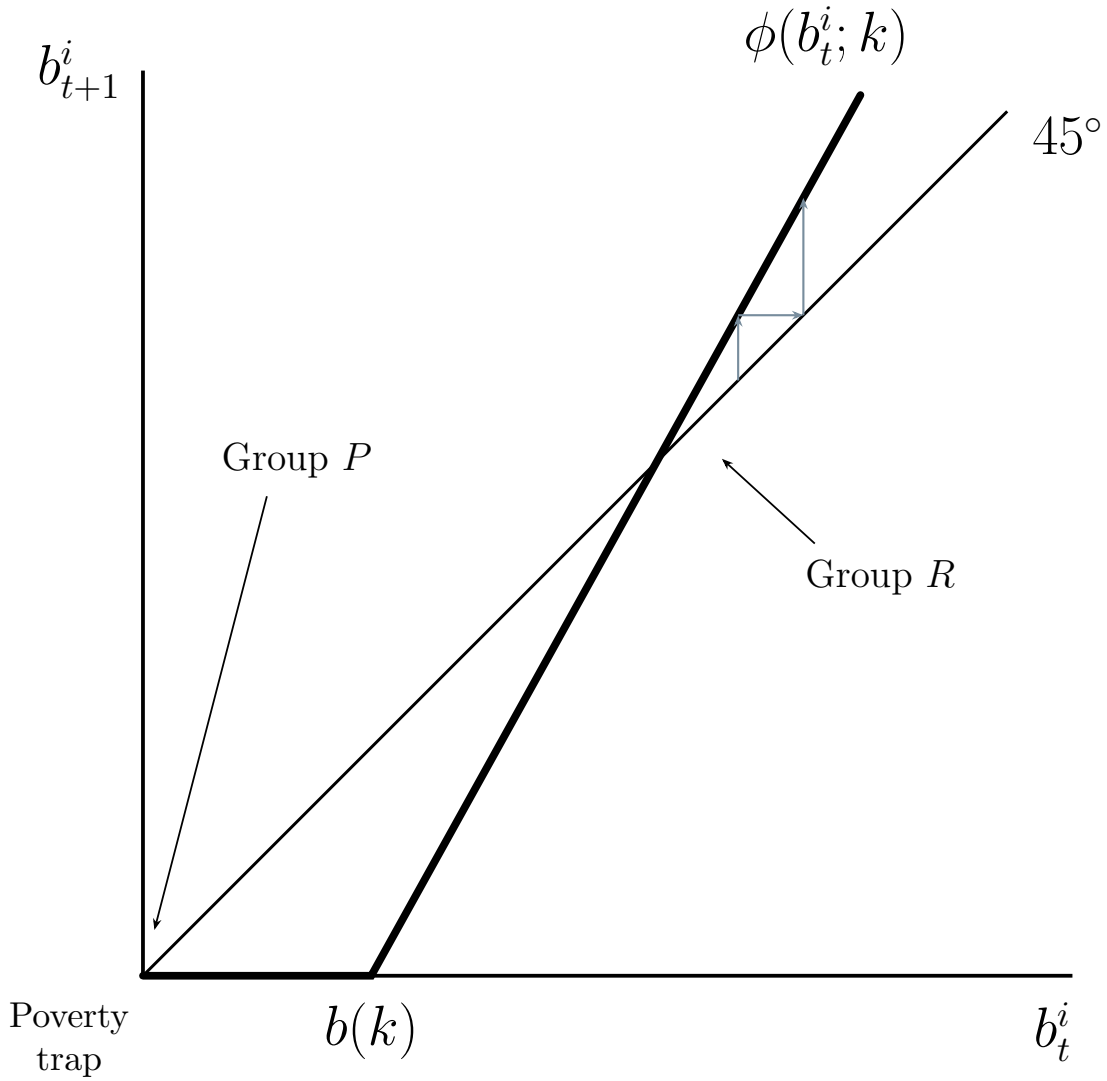
The Process of Development

- Regime I: Physical Capital Accumulation $(k \leq \widetilde{k})$
- Regime II: Human Capital Accumulation $(k > \widetilde{k})$
  - Stage I: $\widetilde{K} < K \leq \widehat{K}$
  - Stage II: $\widehat{K} < K < K^*$
  - Stage III: $K > K^*$

Regime I: Physical Capital Accumulation is the Main Engine of Growth

- Early stages of development $(k \leq \widetilde{k})$
  - Physical capital accumulation is the main engine of growth: $\rho^{HC} \leq \rho^{K}$
  - No investment in education
  - No transfers within Group P
  - Transfers within Group $R \uparrow$
  - Wages $\uparrow$
  - Income inequality $\uparrow$

The Conditional Dynamical System: Regime I



Regime I: The Effect of Inequality

- Inequality fosters economic growth
  - A transfer of wealth from Group R to Group P:
    - Aggregate consumption $\uparrow$
    - Aggregate intergenerational transfers $\downarrow$
    - Rate of capital accumulation $\downarrow$

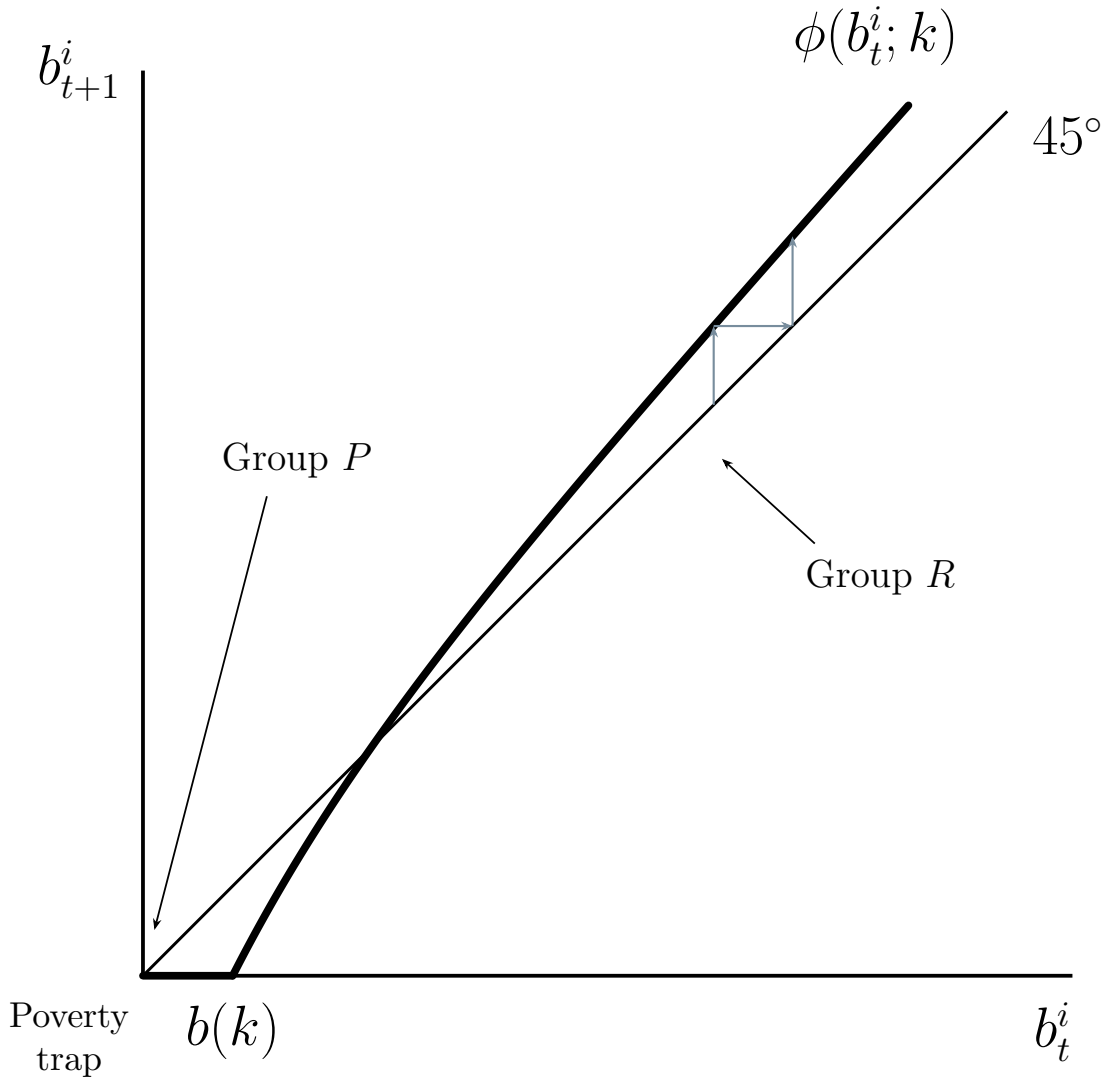
Regime II: Human Capital Accumulation is the Main Engine of Growth

- Mature stages of development: $(k > \widetilde{k})$
  - HC is the engine of growth: $\rho^{HC} \geq \rho^{K}$

Stage I of Regime II: HC Accumulation by group R

- Stage I of Regime II $(\tilde{K} < K \leq \hat{K})$
  - Members of group P
    - No intergenerational transfers
    - No investment in education
  - Members of group R
    - Transfers $\uparrow$
    - Expenditure on education $\uparrow$
  - Wages $\uparrow$
  - Income inequality $\uparrow$

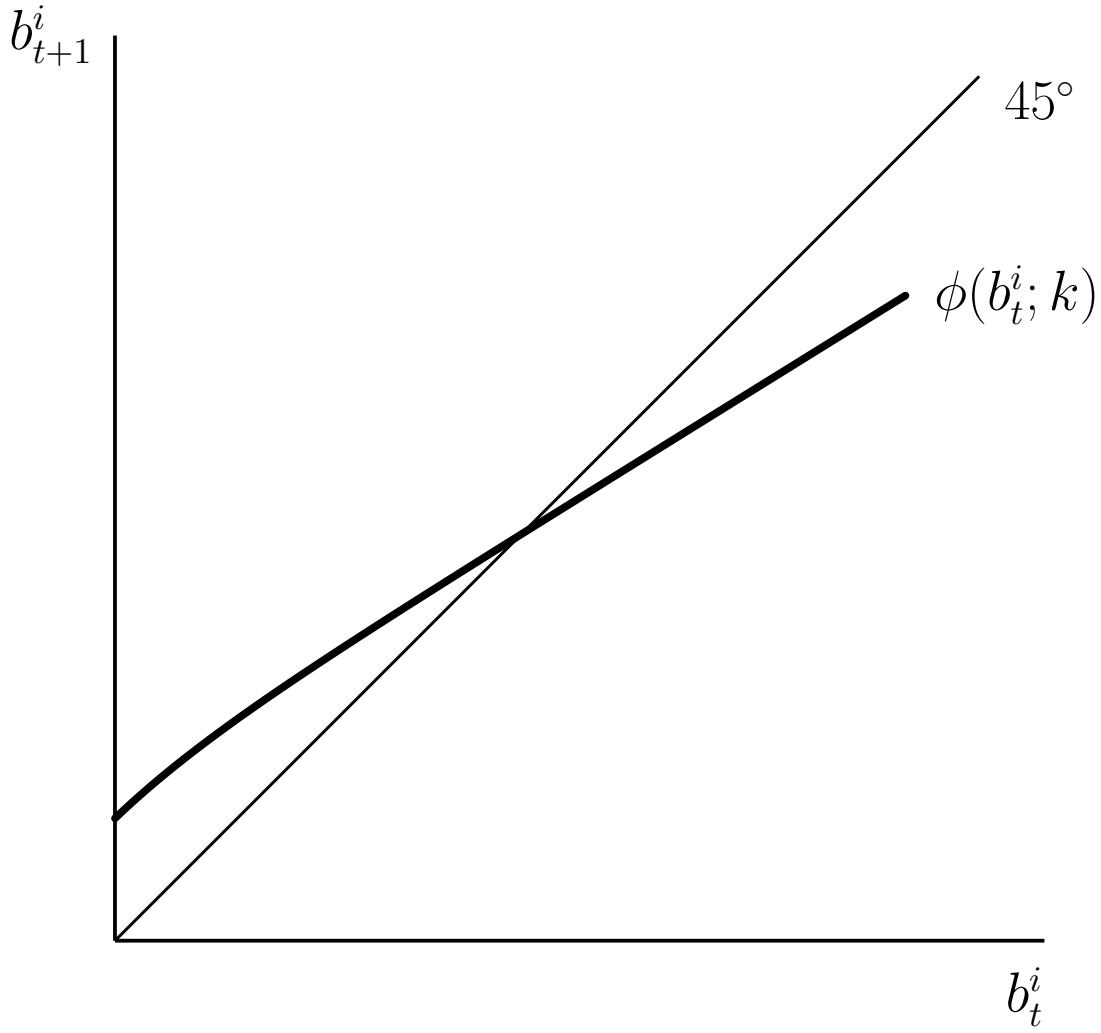
Conditional Dynamical System: Stage I of Regime II



Stage II of Regime II: HC Accumulation by the Poor

- Stage II of Regime II $(\hat{K} < K < K^*)$
  - Members of group P (credit constrained): $\rho^{HC} > \rho^K$
    - Start to make transfers
    - Start to acquire education
  - Members of group R (not credit constrained): $\rho^{HC} = \rho^K$
    - Invest optimally in human and physical capital
  - Greater equality promotes economic development
    - A transfer of wealth from R to P relaxes credit constraints, allowing a more efficient allocation of aggregate investment between HC and PC

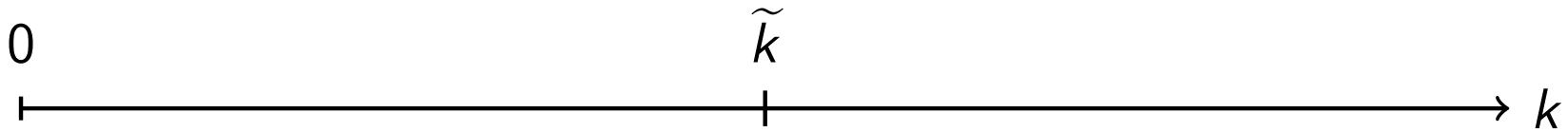
Conditional Dynamical System: Stage II-III of Regime II



Stage III of Regime II : Credit Constraints are not Binding

- All individuals are not credit constrained: $\rho^{HC} = \rho^{K}$
- Inequality has no effect on the process of development

The Changing Role of Inequality in the Development Process



Regime I

$$\rho^K > \rho^{HC}$$

Growth Engine: PC

Inequality fosters growth

Regime II

$$\rho^K \leq \rho^{HC}$$

Growth Engine: HC

Inequality hinders growth

(while credit constraints bind)

Effect of Inequality in Regime II



Stage I	Stage II	Stage III
$\rho^K < (\rho^{HC})^P$	$\rho^K < (\rho^{HC})^P$	$\rho^K = \rho^{HC}$
$\rho^K = (\rho^{HC})^R$	$\rho^K = (\rho^{HC})^R$	
Growth Engines: PC and HC	Growth Engine: HC	Growth Engines: PC and HC

Inequality hinders growth

Testable Implications

- The CMI Approach
  - The effect of inequality on growth depends on the country's level of income
    - Inequality is beneficial for poor economies and harmful for rich ones
- The Unified Approach
  - The effect of inequality on growth depends on the relative return to human and physical capital
    - The higher the relative return to human capital, the more harmful inequality is for economic growth

Implications for DCs and LDCs

- The replacement of physical capital accumulation by human capital accumulation as a prime engine of economic growth has changed the impact of inequality on the process of development
  - Inequality stimulates economic growth in stages of development in which physical capital accumulation is the prime engine of growth
  - Inequality is harmful for economic growth in stages of development in which human capital accumulation is the prime engine of growth
- International capital inflows to LDCs and the adoption of skill-biased technologies may place LDCs directly in the second stage, in which inequality is harmful

References

Main Source:

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